

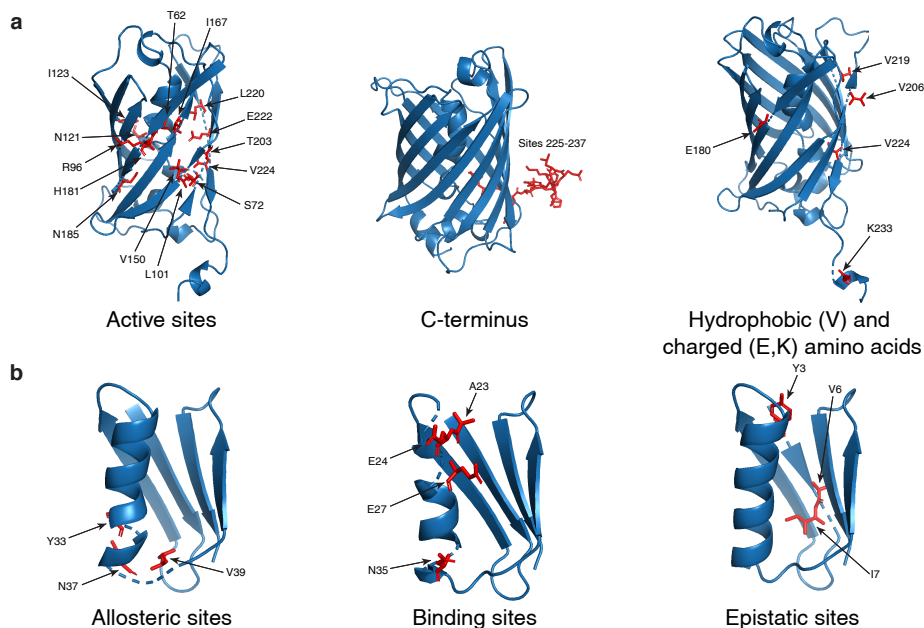


Figure 3: **Analysis of the top-performing steered variants.** **a**, Our analysis of the top-performing GFP variants revealed that steering activated latent features corresponding to key biological motifs, including active site amino acids, the C-terminus, and hydrophobic and charged amino acids. **b**, GB1 variants activated latent features associated with allosteric, binding, and epistatic sites.

To better understand why feature steering designs high-functional variants, we performed a qualitative analysis on the top-performing variants for the green fluorescent protein (GFP) and the IgG-binding domain of protein G (GB1). We identified the ten latent dimensions most strongly associated with changes in fitness. We also analyzed any shifts in their activation patterns between the wildtype and the designed variant. Finally, we projected these activated residues onto the variant’s structure (generated via AlphaFold3 [29]) to identify amino acid concentrations and infer their biological relevance. Further details are provided in Appendix A.6.

Our analysis revealed that SAEs preferentially activated latent features associated with known biological motifs. For instance, in GFP, this includes latents activating on active site amino acids, which are crucial for fluorescence [30], and the C-terminus, a disordered region also known to affect fluorescence (Fig. 3a) [31]. We also found that steering favored latent features corresponding to hydrophobic and charged amino acids, which are essential for maintaining the protein’s structural stability [32]. Similarly, our analysis of the top-performing variants in GB1 highlighted key functional regions (Fig. 3b). The top latents were most active at sites that are allosteric [25], which modulate protein function, or binding, which directly interact with the protein IgG [23]. Furthermore, latents activated on epistatic sites [23], demonstrating the SAE’s ability to design variants in the presence of complex, non-additive mutations. These findings collectively demonstrate that SAEs, even without explicit training, successfully learn and leverage fundamental biological principles to design new variants.

5 Discussion

In this paper, we demonstrated that sparse autoencoders (SAEs) can serve as a powerful tool for low- N tasks. We demonstrated that SAEs consistently outperform their ESM2 counterparts in a variety of low- N fitness extrapolation tasks and are highly effective for generating novel, high-fitness protein variants. Our work expands the biologist’s toolkit for resource-constrained applications and takes the first step toward extracting actionable biological knowledge from pLMs.

Sparsity in SAEs. Although SAEs introduce a larger dimensionality to the linear probes, they consistently outperform their ESM2 counterparts in low- N fitness extrapolation and protein engineering tasks. At first glance, this appears counterintuitive: in low- N regimes, simpler models

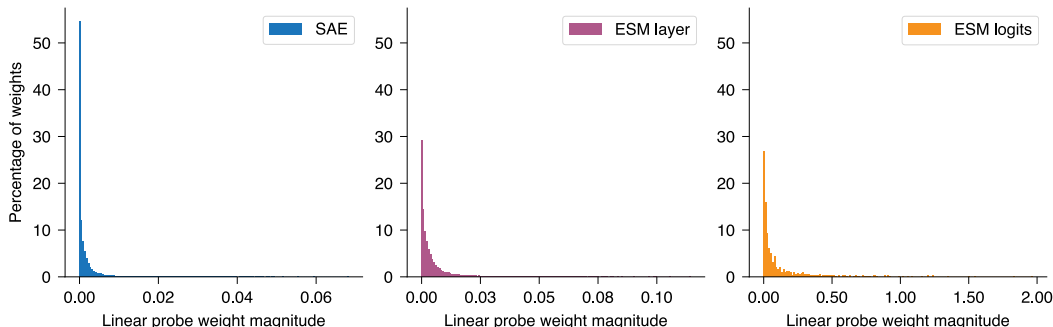


Figure 4: **Sparsity in SAEs underlies improved low- N performance.** Histogram of linear probe weight magnitudes for the SAE latent space, ESM layer embeddings, and ESM logits. Given the top 5% of weights by magnitude, SAE weights explain $37 \pm 9\%$ of the variance, compared to $27 \pm 4\%$ in ESM layer weights and $25 \pm 12\%$ in ESM logits weights. See Appendix A.7 for details on the visualization procedure.

with fewer parameters are typically less prone to overfitting. We attribute the superior performance of SAEs to their ability to compress biologically relevant information into a *sparse* latent space (Fig. 4). To quantify this effect, we measure the variance explained by the magnitude of the top 5% of probe weights. Under this definition, SAE weights explain $38 \pm 9\%$ of the variance, whereas ESM layer and ESM logits weights explain $28 \pm 3\%$ and $31 \pm 17\%$, respectively. These results suggest that SAEs compress information from ESM2 into a more compact and disentangled representation, where the biological relevant signal is concentrated in a select few latents. In the low- N regime, this compression is particularly advantageous: sparser models are less prone to overfitting and thus generalize more effectively from limited experimental data. This also allows each high-impact latent to disentangle which amino acids contribute to fitness, enhancing the effectiveness of our steering approach.

Low- N Performance Variability. Across fitness extrapolation tasks, we observe relatively high standard deviation in all methods. This is not surprising, given that the linear probes are trained in the low- N regime, making the performance sensitive to which sequences are sampled. Despite this variability, we emphasize that SAEs outperform their ESM2 counterparts in 58% of cases on average, indicating a consistent advantage. Moreover, SAEs tend to design more high-functioning variants, suggesting that their sparse latent space captures a more informative view of the functional landscape.

Limitations and Future Work. Our work analyzes the performance of SAEs across a wide range of proteins and molecular functions. However, our evaluation could be extended to other proteins with clinically relevant functions such as antibiotic resistance and viral replication, which may open new avenues for therapeutic design. We also observed that SAE performance is strongly influenced by the number of MSA sequences available for training. For example, proteins such as DLG4, GRB2, and F7YBW8, which consistently achieved high fitness extrapolation correlations, each had more than 20,000 MSA sequences. Future work should explore strategies for robust SAE training when MSAs are shallow or unavailable.

To validate the designed variants, our protein engineering experiments rely on trained fitness models *in silico*. While this is a good proxy for fitness, further validation through wet lab experiments is necessary to verify the function of designed variants. Nevertheless, our results on the SPG1_STRSG_Wu DMS assay, for which ground-truth fitness values are available for all combinatorial variants, demonstrate that our SAE steering approach still produces the highest-fitness variants compared to other methods. Additionally, we clarify that successful protein design requires not only highly-functional, but also highly-stable variants. Certain mutations that promote function may also destabilize the protein [33, 34], reinforcing the need for wet lab experiments to test designs. A promising future direction is to couple SAE steering with physics-based tools such as Rosetta [35] to jointly optimize for both function and stability.

Lastly, in our protein engineering experiments, we constrained variants to be a maximum of five mutations away from the wildtype. Beyond this radius, we found it difficult to design highly-functional sequences using just one predictive latent. Additionally, because predictive information

is concentrated in a small number of latents, we restricted the amount of variants designed to 50 per DMS assay. Future work towards expanding the design space could involve steering multiple latents at once, which could enable both further mutational exploration and more diverse pools of functional variants. However, we note that in silico evaluation tools become less reliable the further away variants from the wildtype are, reinforcing the need for wet lab validation.

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